

Problem 1

If $\underline{B}(x,y,z) = xy^2 \underline{\hat{x}} + xy^2z^3 \underline{\hat{y}} + x^2yz^2 \underline{\hat{z}}$

find $\nabla \times \underline{B}$, $\nabla \cdot \underline{B}$.

Solution:

$$B_x = xy^2, \quad B_y = xy^2z^3, \quad B_z = x^2yz^2$$

Divergence

$$\nabla \cdot \underline{B} = \frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z}$$

$$= y^2 + xz^3 \cdot 2y + x^2y \cdot 2z$$

$$= y^2 + 2xyz^3 + 2x^2yz$$

Curl:

$$\nabla \times \underline{B} = \begin{vmatrix} \underline{\hat{x}} & \underline{\hat{y}} & \underline{\hat{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2 & xy^2z^3 & x^2yz^2 \end{vmatrix}$$

$$= \underline{\hat{x}} \begin{vmatrix} \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ xy^2z^3 & x^2yz^2 \end{vmatrix} - \underline{\hat{y}} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial z} \\ xy^2 & x^2yz^2 \end{vmatrix} + \underline{\hat{z}} \begin{vmatrix} \frac{\partial}{\partial x} & \frac{\partial}{\partial y} \\ xy^2 & xy^2z^3 \end{vmatrix}$$

$$\begin{aligned}
 &= \hat{x} (x^2z^2 - xy^2z^2) - \hat{y} (2xy^2z - 0) + \hat{z} (y^2z^3 - xy) \\
 &= (x^2z^2 - 3xy^2z^2) \hat{x} - 2xy^2z \hat{y} + (y^2z^3 - xy) \hat{z}
 \end{aligned}$$

Problem 2.

prove that for an arbitrary vector $\underline{A}(x, y, z)$ and an arbitrary scalar function $\phi(x, y, z)$

$$\nabla \cdot (\nabla \times \underline{A}) = 0$$

$$\nabla \times (\nabla \phi) = \underline{0}$$

Scalar function of (x, y, z)
Ex $A_z = x^2 + z$

Proof.

$$\text{Let } \underline{A} = \underline{A_x} \hat{x} + \underline{A_y} \hat{y} + \underline{A_z} \hat{z}$$

By definition

$$\text{First } \nabla \times \underline{A} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

$$= \hat{x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) - \hat{y} \left(\frac{\partial A_z}{\partial x} - \frac{\partial A_x}{\partial z} \right) + \hat{z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$= \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) \hat{x} + \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) \hat{y} + \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \hat{z}$$

Second.

$$\nabla \cdot (\nabla \times \underline{A}) = \frac{\partial}{\partial x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right)$$

$$\begin{aligned} \nabla \cdot (\nabla \times \underline{A}) &= \frac{\partial}{\partial x} \left(\frac{\partial A_z}{\partial y} - \frac{\partial A_y}{\partial z} \right) + \frac{\partial}{\partial y} \left(\frac{\partial A_x}{\partial z} - \frac{\partial A_z}{\partial x} \right) + \frac{\partial}{\partial z} \left(\frac{\partial A_y}{\partial x} - \frac{\partial A_x}{\partial y} \right) \\ &= \cancel{\frac{\partial^2 A_z}{\partial x \partial y}} - \frac{\partial^2 A_y}{\partial x \partial z} + \frac{\partial^2 A_x}{\partial y \partial z} - \cancel{\frac{\partial^2 A_z}{\partial y \partial x}} + \frac{\partial^2 A_z}{\partial x \partial z} - \frac{\partial^2 A_y}{\partial z \partial y} \end{aligned}$$

Note $\frac{\partial^2 A_z}{\partial x \partial y} = \frac{\partial^2 A_z}{\partial y \partial x}$.

Since x, y are orthogonal
so does each other Her -

$$\therefore \nabla \cdot (\nabla \times \underline{A}) = 0$$

proof done.

ϕ

proof of $\nabla \times (\nabla \phi) = 0$

Note, ϕ is a scalar function $\phi(x, y, z)$

$$\therefore \nabla \phi = \frac{\partial \phi}{\partial x} \underline{\hat{x}} + \frac{\partial \phi}{\partial y} \underline{\hat{y}} + \frac{\partial \phi}{\partial z} \underline{\hat{z}}$$

$$\therefore \nabla \times (\nabla \phi) = \begin{vmatrix} \underline{\hat{x}} & \underline{\hat{y}} & \underline{\hat{z}} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$\uparrow$ Vector $\underbrace{\hspace{1cm}}$ Vector

$$= \underline{\hat{x}} \left(\frac{\partial^2 \phi}{\partial y \partial z} - \frac{\partial^2 \phi}{\partial z \partial y} \right) - \underline{\hat{y}} \left(\frac{\partial^2 \phi}{\partial x \partial z} - \frac{\partial^2 \phi}{\partial z \partial x} \right) + \underline{\hat{z}} \left(\frac{\partial^2 \phi}{\partial x \partial y} - \frac{\partial^2 \phi}{\partial y \partial x} \right)$$

$$= \underline{\hat{x}} \cdot 0 + \underline{\hat{y}} \cdot 0 + \underline{\hat{z}} \cdot 0$$

$$= \underline{0}$$

Note, the curl is a vector.

problem 3.

The vector Laplacian is defined as

$$\nabla^2 \underline{V} \equiv \nabla(\nabla \cdot \underline{V}) - \nabla \times (\nabla \times \underline{V})$$

prove in rectangular coordinates:

$$\nabla = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$$

$$\nabla^2 \underline{V} = \hat{x} \nabla^2 V_x + \hat{y} \nabla^2 V_y + \hat{z} \nabla^2 V_z$$

$$\underline{V} = V_x \hat{x} + V_y \hat{y} + V_z \hat{z}$$

$$\nabla \cdot \underline{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$$

$$\therefore \nabla(\nabla \cdot \underline{V}) = \frac{\partial}{\partial x} \left(\frac{\partial V_x}{\partial x} \right) \hat{x} + \frac{\partial}{\partial y} \left(\frac{\partial V_y}{\partial y} \right) \hat{y} + \frac{\partial}{\partial z} \left(\frac{\partial V_z}{\partial z} \right) \hat{z}$$

$$= \frac{\partial^2 V_x}{\partial x^2} \hat{x} + \frac{\partial^2 V_y}{\partial y^2} \hat{y} + \frac{\partial^2 V_z}{\partial z^2} \hat{z}$$

$$\nabla \times \underline{V} = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix}$$

$$= \hat{x} \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) + \hat{y} \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) + \hat{z} \left(\frac{\partial V_z}{\partial x} - \frac{\partial V_x}{\partial y} \right)$$

$$\therefore \nabla \times (\nabla \times V) = \begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) & \left(\frac{\partial V_x}{\partial z} - \frac{\partial V_z}{\partial x} \right) & \left(\frac{\partial V_z}{\partial x} - \frac{\partial V_x}{\partial y} \right) \end{vmatrix}$$

$$= \hat{x} \left(\frac{\partial}{\partial y} \left(\frac{\partial V_z}{\partial x} - \frac{\partial V_x}{\partial z} \right) - \frac{\partial}{\partial z} \left(\frac{\partial V_z}{\partial y} - \frac{\partial V_y}{\partial z} \right) \right) + \boxed{} \hat{y} + \boxed{} \hat{z}$$

$$= \hat{x} \left(\frac{\partial^2 V_z}{\partial y \partial x} - \frac{\partial^2 V_x}{\partial y \partial z} - \frac{\partial^2 V_z}{\partial z \partial y} + \frac{\partial^2 V_y}{\partial z^2} \right)$$

Substitution, let's focus on $\hat{x}$ components

$$\nabla(\nabla \cdot V)_x = \frac{\partial^2 V_x}{\partial x^2} \hat{x} + \boxed{} \hat{y} + \boxed{} \hat{z}$$